Generalizing Dataset Distillation via Deep Generative Prior Supplemental Material

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A. StyleGAN Latent Spaces

Here we further elaborate on our embedding spaces for those unfamiliar with the StyleGAN architecture. Some details will be left out for easier digestion. Visualizations of these embedding spaces can be seen in Figure 1.

For standard image generation with StyleGAN 3 [3] (and other StyleGAN models), a random vector is first sampled from the multi-variate standard normal distribution: $z \sim \mathcal{N}(0, I)$. This random Gaussian vector is then fed through a "mapping" network (typically a simple MLP) to obtain a "style code" W. In a class-conditional StyleGAN, this "mapping" network is the *only* place where the class information is used. This W is then passed to every "style block" of the "synthesis" network as used to modulate the convolutional layers of each block. The "synthesis" network takes in a learned constant as input and uses the modulated convolutions of the style blocks to generate the realistic image.

For each distilled sample in W^+ space, we optimize a *different* W code for each style block and use the synthesis network to generate our synthetic data. The "mapping" network is not used aside from initializing W. In Fn space, we *directly* optimize the feed-forward input to the nth style block as well as the W codes for all subsequent style blocks. Any preceding style blocks and W codes are simply ignored.

B. Dataset Specifications

Our high-resolution data is taken directly from the ImageNet 1k dataset [2] using PyTorch's built-in ImageNet loader [4]. To train our expert trajectories, we use data from the ImageNet training set. To compile *our* training set for the expert trajectories, we select the classes from the given subset, resize the short side of the image to the given resolution, and take a center crop according to the given resolution (as done by `MTT` [1]). The validation set is obtained in the same way from the ImageNet validation set.

For an enumeration of which ImageNet classes are in each of our datasets, please see 1.

References

[1] George Cazenavette, Tongzhou Wang, Antonio Torralba, Alexei A. Efros, and Jun-Yan Zhu. Dataset distillation by matching training trajectories. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. 1

[2] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *CVPR*, 2009. 1

[3] Tero Karras, Miika Aittala, Samuli Laine, Erik Härkönen, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. Alias-free generative adversarial networks. In *Proc. NeurIPS*, 2021. 1

[4] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. Pytorch: An imperative style, high-performance deep learning library. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems 32*, pages 8024–8035. Curran Associates, Inc., 2019. 1

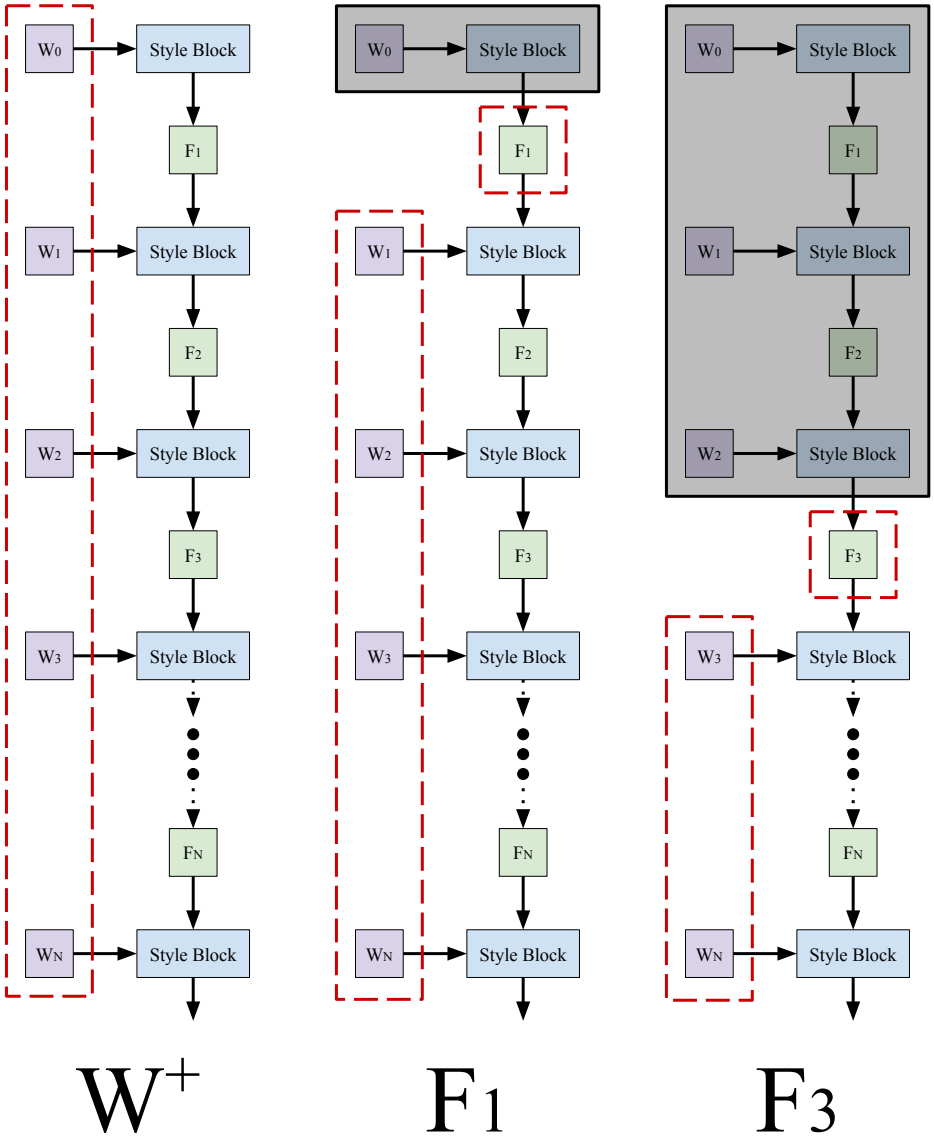


Figure 1. Different optimization spaces of StyleGAN. Latent variables boxed in red are directly optimized while those that are grayed out are not used at all. Note: the "mapping" network is omitted here since we do not use it in any of our optimization spaces.

Dataset	0	1	2	3	4	5	6	7	8	9
ImageNet-A	Leonberg	Probiscis Monkey	Rapeseed	Three-Toed Sloth	Cliff Dwelling	Yellow Lady's Slipper	Hamster	Gondola	Orca	Limpkin
ImageNet-B	Spoonbill	Website	Lorikeet	Hyena	Earthstar	Trollybus	Echidna	Pomeranian	Odometer	Ruddy Turnstone
ImageNet-C	Freight Car	Hummingbird	Fireboat	Disk Brake	Bee Eater	Rock Beauty	Lion	European Gallinule	Cabbage Butterfly	Goldfinch
ImageNet-D	Ostrich	Samoyed	Snowbird	Brabancon Griffon	Chickadee	Sorrel	Admiral	Great Gray Owl	Hornbill	Ringlet
ImageNet-E	Axolotl	Tree Frog	King Snake	American Chameleon	Iguana	Eft	Fire Salamander	Box Turtle	American Alligator	Agama
ImageNette	Tench	English Springer	Cassette Player	Chainsaw	Church	French Horn	Garbage Truck	Gas Pump	Golf Ball	Parachute
ImageWoof	Australian Terrier	Border Terrier	Samoyed	Beagle	Shih-Tzu	English Foxhound	Rhodesian Ridgeback	Dingo	Golden Retriever	English Sheepdog
ImageNet-Birds	Peacock	Flamingo	Macaw	Pelican	King Penguin	Bald Eagle	Toucan	Ostrich	Black Swan	Cockatoo
ImageNet-Fruits	Pineapple	Banana	Strawberry	Orange	Lemon	Pomegranate	Fig	Bell Pepper	Cucumber	Green Apple
ImageNet-Cats	Tabby Cat	Bengal Cat	Persian Cat	Siamese Cat	Egyptian Cat	Lion	Tiger	Jaguar	Snow Leopard	Lynx

Table 1. Class listings for our ImageNet subsets. Visualizations show classes in the same order given here.